# The Æther-flow Interpretation of Relativity

Substrate Kinematics, Candidate Dynamics, and the Derivational Bridge

**Abstract**

This manuscript states the next substantive step beyond the current exact-closure sequence of Æther-flow Interpretation of Relativity. Its purpose is not to claim that Einsteinian gravity has already been derived from substrate microphysics. Its purpose is narrower and more disciplined: to make explicit the minimum substrate kinematics and one explicit candidate dynamical class needed for a future derivational program.

The paper therefore introduces a four-dimensional substrate manifold $\mathcal{A}$ equipped with a positive-definite substrate metric $G_{AB}$, a unit ordered-flow field $U^A$, an order scalar $S(X)$, and additional substrate variables $Q$. It then defines the local experiential three-space by the projector orthogonal to $U^A$, constructs a Lorentzian bridge metric $\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$, and uses a local observer map to recover the effective relativistic metric $g_{\mu\nu}$ seen by embedded observers. On that bridge, it proposes a candidate substrate action class, a universal matter-coupling route through $g_{\mu\nu}$, a first mode-control regime for the extra substrate sector, and an explicit exact-closure test.

The resulting claim is candidate-dynamical rather than fully derived. The paper states the missing bridge from ontology to explicit substrate structure, shows how observed three-dimensional space and S-time can be anchored in that structure, and records the conditions any completed substrate action must satisfy if it is to recover the exact Einsteinian closure already adopted elsewhere in the active sequence. Exact closure therefore remains the benchmark and reversion theory until those stronger derivational conditions are actually met.

## 1. Introduction

The active manuscript sequence of Æther-flow Interpretation of Relativity now fixes the ontology, the exact effective dynamics, the consistency conditions, the relativistic recovery statement, and the congruence-based flow geometry of the exact-closure theory. What it still lacks is the deeper bridge from ontology to explicit substrate kinematics and derivation.

That missing step is the subject of the present paper. The goal is not another GR / SR recovery statement. Nor is the goal to replace the exact Einsteinian closure already adopted in the active theory line. The goal is to define the minimum substrate-level data and one explicit candidate dynamical route from which a future derivation would have to begin.

**Framework and claim boundary.** Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of

Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present manuscript stays strictly within that claim boundary. It does not claim a completed recovery of the Einstein equations from substrate variables. It does not present a finished microphysical action that has already passed a full consistency analysis. It does make one positive move: it replaces the previously schematic phrase "future substrate manuscript" with an explicit bridge architecture, a candidate substrate action class, and success conditions that can be stated cleanly.

## 2. Role of the Derivational-Bridge Step

The foundations manuscript already introduced a four-dimensional substrate manifold $\mathcal{A}$, an ordered-motion map $\Phi_\lambda$, and the idea that observed three-dimensional space is a local experiential slice of the deeper substrate. The dynamics manuscript then fixed the exact effective law of Æther-flow Interpretation of Relativity to be Einsteinian. The flow-geometry manuscript identified the Æther-flow with a congruence-based interpretation of that exact relativistic geometry.

What is still missing between those layers is the explicit substrate side of the bridge. In particular, the theory still requires:

1. a disciplined specification of the substrate variables from which a relativistic metric sector could emerge;
2. an explicit local bridge from those substrate variables to the effective Lorentzian geometry seen by observers;
3. a clear statement of what counts as a successful derivation and what would still count only as an unfinished ansatz.

The present manuscript addresses those three points. It should therefore be read as a bridge-and-candidate-dynamics paper, not as a completed derivation paper.

## 3. Minimal Substrate Kinematics

**Definition 1** (Minimal substrate data)**.** *The minimal kinematic data of the derivational bridge are the tuple*

$$(\mathcal{A}, G_{AB}, U^A, S, Q),$$

*where:*

1. *$\mathcal{A}$ is a smooth four-dimensional substrate manifold;*
2. *$G_{AB}$ is a positive-definite substrate metric on $\mathcal{A}$;*
3. *$U^A$ is a smooth unit ordered-flow field satisfying*

$$G_{AB}U^AU^B = 1; \tag{1}$$

*4. $S : \mathcal{A} \to \mathbb{R}$ is an order scalar satisfying*

$$U^A \partial_A S > 0; \tag{2}$$

*5. $Q$ denotes any additional substrate variables needed for a later completed dynamics.*

Equation (1) makes $U^A$ the explicit kinematic carrier of Æther-flow at the substrate level. Equation (2) makes $S$ an order variable that increases along the ordered motion.

This separation between $U^A$ and $S$ is important. The ordered-flow field and the order scalar should not be collapsed automatically into one object. If they were identified too early, the framework could accidentally forbid the rotational and vorticity sectors already needed in the relativistic flow-geometry manuscript.

The ordered-motion map introduced in the foundations manuscript is now made explicit through the integral curves of $U^A$:

$$\frac{\mathrm{d}X^A}{\mathrm{d}\lambda} = U^A(X(\lambda)), \qquad \Phi_\lambda(X_0) = X(\lambda). \tag{3}$$

Thus $\Phi_\lambda$ is the substrate transport generated by $U^A$, while $S$ provides a monotone ordering of that transport.

**Remark 1.** *At this stage $U^A$ is a substrate variable, not yet an independently observed low-energy preferred-frame field. The active exact-closure line is preserved precisely because the present manuscript defines only the bridge architecture, not a surviving extra low-energy sector.*

## 4. Local Experiential Space and S-Time

Given the unit flow field $U^A$, define the projector orthogonal to the ordered motion by

$$P_{AB} = G_{AB} - U_A U_B, \tag{4}$$

where $U_A = G_{AB} U^B$.

**Proposition 1.** *The tensor $P_{AB}$ is positive semidefinite on $T\mathcal{A}$, has kernel spanned by $U^A$, and is positive definite on the three-dimensional subspace orthogonal to $U^A$.*

*Proof.* For any tangent vector $V^A$, decompose

$$V^A = \alpha U^A + W^A, \qquad U_A W^A = 0.$$

Then

$$P_{AB} V^A V^B = G_{AB} W^A W^B \geq 0,$$

with equality if and only if $W^A = 0$, that is, if $V^A$ is proportional to $U^A$. Since $G_{AB}$ is positive definite, its restriction to the orthogonal complement of $U^A$ is positive definite. The orthogonal complement is three-dimensional because $\dim \mathcal{A} = 4$. Therefore $P_{AB}$ has the stated properties. □

Equation (4) is the local substrate realization of observed three-dimensional space as a local experiential slice. The word *local* matters. The projector $P_{AB}$ always defines a local three-space

orthogonal to the ordered motion, but those local spaces need not integrate into a single global foliation.

To see the distinction, define the projected antisymmetric derivative

$$\Omega_{AB} \equiv P_A{}^C P_B{}^D \nabla_{[C} U_{D]}, \tag{5}$$

where $\nabla_A$ is the Levi-Civita connection of $G_{AB}$.

**Proposition 2.** *The local experiential three-spaces orthogonal to $U^A$ integrate to hypersurfaces if and only if*

$$U_{[A} \nabla_B U_{C]} = 0, \tag{6}$$

*equivalently if and only if $\Omega_{AB} = 0$.*

*Proof.* This is the Frobenius criterion applied to the one-form $U_A$. Hypersurface orthogonality is equivalent to $U_{[A} \nabla_B U_{C]} = 0$. Because $\Omega_{AB}$ is the orthogonally projected antisymmetric derivative, it vanishes exactly when the orthogonal distribution defined by $P_{AB}$ is integrable. □

This result is structurally important. It shows that the phrase *local experiential slice* should not be read as a promise of a universal global slicing in every regime. In particular, the derivational bridge remains compatible with the later rotational and frame-dragging sectors, which require nonzero vorticity in appropriate situations.

The order scalar $S$ then provides the minimal explicit anchor for S-time. A simple kinematic class of S-time functionals is

$$\mathcal{S}[\Gamma, \Psi] = \int_\Gamma \rho(X, \Psi)\, dS, \qquad \rho(X, \Psi) > 0, \tag{7}$$

where $\Gamma$ is an admissible observer history and $\Psi$ denotes the physical state registered along that history.

For a parameterized admissible history $X^A(\lambda)$, (7) gives

$$\frac{\mathrm{d}\mathcal{S}}{\mathrm{d}\lambda} = \rho(X(\lambda), \Psi(\lambda))\, \frac{\mathrm{d}S}{\mathrm{d}\lambda}. \tag{8}$$

Whenever $dS/d\lambda > 0$, the S-time functional is monotone. This is the explicit bridge between the substrate order variable and the experienced order of change.

## 5. Kinematic Bridge to a Lorentzian Metric

The next step is the central bridge ansatz of the paper.

**Definition 2** (Bridge metric)**.** *Given the substrate metric $G_{AB}$ and the unit ordered-flow field $U^A$, define the bridge metric*

$$\mathfrak{g}_{AB} = P_{AB} - U_A U_B = G_{AB} - 2U_A U_B. \tag{9}$$

**Proposition 3.** *The bridge metric $\mathfrak{g}_{AB}$ has Lorentzian signature $(-,+,+,+)$.*

*Proof.* For any $V^A = \alpha U^A + W^A$ with $U_A W^A = 0$,

$$\mathfrak{g}_{AB} V^A V^B = -\alpha^2 + P_{AB} W^A W^B. \tag{10}$$

By the previous proposition, $P_{AB}$ is positive definite on the three-dimensional space orthogonal to $U^A$. Therefore (10) has one negative direction, carried by $U^A$, and three positive directions, carried by the orthogonal complement. Hence $\mathfrak{g}_{AB}$ is Lorentzian with signature $(-,+,+,+)$. □

Equation (10) already gives the causal classification induced by the bridge:

1. $V^A$ is timelike if $\alpha^2 > P_{AB}W^AW^B$;
2. $V^A$ is null if $\alpha^2 = P_{AB}W^AW^B$;
3. $V^A$ is spacelike if $\alpha^2 < P_{AB}W^AW^B$.

This is the first precise statement of how a Lorentzian observer-accessible sector can arise from explicit substrate kinematics while the underlying substrate metric remains positive definite.

Define the corresponding effective four-velocity field on the substrate by

$$\mathfrak{u}^A = c\,U^A. \tag{11}$$

Then

$$\mathfrak{g}_{AB}\,\mathfrak{u}^A\mathfrak{u}^B = -c^2, \tag{12}$$

and the $\mathfrak{g}$-rest-space projector is

$$\mathfrak{g}_{AB} + \frac{1}{c^2}\mathfrak{u}_A\mathfrak{u}_B = P_{AB}. \tag{13}$$

Equation (13) is the key bridge identity. It shows that the local experiential three-space determined on the substrate side is exactly the same object as the observer rest-space geometry defined after the Lorentzian bridge is introduced.

## 6. Observer Map and the Effective Relativistic Sector

To connect the bridge metric to the effective relativistic variables used in the rest of the manuscript sequence, introduce a local observer map

$$\Xi : M \to \mathcal{A},$$

where $M$ is the observer-accessible spacetime manifold and $\Xi$ is taken locally to be a diffeomorphic identification.

The effective spacetime metric is then defined by pullback:

$$g_{\mu\nu} = \Xi^*\mathfrak{g}_{AB} = \frac{\partial X^A}{\partial x^\mu}\frac{\partial X^B}{\partial x^\nu}\mathfrak{g}_{AB}\big(X(x)\big). \tag{14}$$

Likewise the effective observer field is

$$u^\mu = (\Xi^{-1})_*\mathfrak{u}^A. \tag{15}$$

**Proposition 4.** *The local rest-space metric seen by observers in the effective relativistic sector is the pullback of the substrate projector:*

$$h_{\mu\nu} \equiv g_{\mu\nu} + \frac{1}{c^2}u_\mu u_\nu = \Xi^* P_{AB}. \tag{16}$$

*Proof.* By (14) and (15), the pair $(g_{\mu\nu}, u^\mu)$ is the image of $(\mathfrak{g}_{AB}, \mathfrak{u}^A)$ under the same local identification $\Xi$. Pulling back (13) therefore gives

$$\Xi^*\left(\mathfrak{g}_{AB} + \frac{1}{c^2}\mathfrak{u}_A\mathfrak{u}_B\right) = \Xi^* P_{AB},$$

which is exactly (16). □

This proposition is the cleanest statement of the derivational bridge. Observed three-dimensional space as a local experiential slice is not left as metaphor. It is identified explicitly with the pullback of the substrate rest-space projector orthogonal to the ordered flow.

The standard operational relativistic relations then follow on the effective side:

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu}\, dx^\mu dx^\nu = -\frac{1}{c^2}\mathfrak{g}_{AB}\, dX^A dX^B, \tag{17}$$

and for null propagation,

$$0 = g_{\mu\nu}\, dx^\mu dx^\nu = \mathfrak{g}_{AB}\, dX^A dX^B. \tag{18}$$

**Remark 2.** *The congruence-based flow geometry developed elsewhere in the active sequence now acquires a substrate-level origin. After the bridge map is made, the effective observer field $u^\mu$ is the relativistic image of the deeper ordered-flow field $U^A$. Expansion, acceleration, vorticity, and shear in the effective geometry are therefore not inserted by hand; they are the observer-accessible kinematic face of the deeper substrate motion.*

## 7. Candidate Substrate Dynamics

The bridge is not yet scientifically useful unless one can state at least one explicit candidate dynamics built on it. The purpose of the present section is therefore not to prove emergence, but to replace a purely schematic future target by a concrete action class that can later be tested.

### 7.1. Candidate Action Class

**Proposition 5.** *If $G_{AB}U^AU^B = 1$, then the bridge metric satisfies*

$$\det \mathfrak{g}_{AB} = -\det G_{AB}. \tag{19}$$

*Proof.* Choose a $G$-orthonormal basis adapted to $U^A$. In that basis,

$$G_{AB} = \mathrm{diag}(1,1,1,1), \qquad \mathfrak{g}_{AB} = \mathrm{diag}(-1,1,1,1).$$

Therefore $\det \mathfrak{g}_{AB} = -1 = -\det G_{AB}$. Since both determinants are basis-invariant, (19) holds generally. □

Equation (19) implies $\sqrt{-\det \mathfrak{g}} = \sqrt{G}$, so the same invariant volume element may be written using either the positive-definite substrate metric or the induced bridge metric.

For concreteness, let the extra substrate sector be a multiplet $Q^I$, $I = 1, \ldots, N$, with positive-definite mass matrix $M_{IJ}$. A candidate substrate action class compatible with the bridge is then

$$S_{\text{cand}} = \frac{c^3}{16\pi G_N} \int_{\mathcal{A}} d^4X \sqrt{G}\,(R[\mathfrak{g}] - 2\Lambda_{\mathcal{A}}) + \int_{\mathcal{A}} d^4X \sqrt{G}\,\mathcal{L}_{\text{aux}} + S_{\text{matter}}[\Xi^*\mathfrak{g}, \psi], \tag{20}$$

where

$$\begin{aligned}\mathcal{L}_{\text{aux}} = {} & \lambda_U (G_{AB}U^A U^B - 1) - \frac{\kappa_S}{2} P^{AB}\partial_A S\,\partial_B S - \frac{m_S^2}{2}\big(U^A\partial_A S - \sigma_0\big)^2 - V_S(S) \\ & - \frac{1}{2}\delta_{IJ}P^{AB}(D_A Q^I)(D_B Q^J) - \frac{1}{2}M_{IJ}Q^I Q^J - W(S,Q) \\ & - \frac{\kappa_\theta}{2}\big(\nabla_A U^A\big)^2 - \frac{\kappa_\Omega}{4}\Omega_{AB}\Omega^{AB}.\end{aligned} \tag{21}$$

Here $\lambda_U$ is a Lagrange multiplier enforcing (1), $\sigma_0 > 0$ sets the order scale along the flow, $D_A$ is any internal covariant derivative acting on the $Q^I$, and all contractions are performed with $G_{AB}$ and its inverse.

The first term in (20) is written directly in terms of the bridge Ricci scalar $R[\mathfrak{g}]$ for a clear reason: it makes the candidate model answerable to the exact Einsteinian benchmark already fixed elsewhere in the active sequence. This is still not a completed derivation. The Einsteinian bridge curvature has been posited as the candidate leading bridge-sector dynamics, not recovered from a deeper still-hidden layer.

### 7.2. Candidate Equations and Observer-Accessible Metric Equation

Variation of (20) with respect to $\lambda_U$ reproduces the unit-flow constraint (1). Variation with respect to the order scalar gives

$$\kappa_S \nabla_A\Big(P^{AB}\partial_B S\Big) + m_S^2 \nabla_A\Big[(U^C\partial_C S - \sigma_0)U^A\Big] - V_S'(S) - \frac{\partial W}{\partial S} = 0. \tag{22}$$

Variation with respect to the extra fields $Q^I$ gives

$$\nabla_A\Big(P^{AB}D_B Q_I\Big) - M_{IJ}Q^J - \frac{\partial W}{\partial Q^I} = 0, \tag{23}$$

where $Q_I = \delta_{IJ}Q^J$.

The full $G_{AB}$ and $U^A$ Euler–Lagrange equations are lengthy because $\mathfrak{g}_{AB}$, $P_{AB}$, and $\Omega_{AB}$ all depend on them. For the purpose of the present manuscript, the decisive object is instead the observer-accessible pullback of the bridge-sector equation:

$$G_{\mu\nu}[g] + \Lambda_{\mathcal{A}} g_{\mu\nu} = \frac{8\pi G_N}{c^4} T_{\mu\nu} + \Delta_{\mu\nu}^{(U,S,Q)}. \tag{24}$$

Here $T_{\mu\nu}$ is the stress tensor derived from $S_{\text{matter}}[g, \psi]$, while $\Delta_{\mu\nu}^{(U,S,Q)}$ collects the pullback of all non-Einsteinian bridge corrections generated by the auxiliary substrate sector. Equation (24) is the correct place to formulate the exact-closure test: the candidate model succeeds only if $\Delta_{\mu\nu}^{(U,S,Q)}$ vanishes or is driven below observer-accessible relevance in the intended regime.

## 8. Universal Matter Coupling Through the Bridge

The matter-coupling route must be stated explicitly, because exact closure depends on it as much as the gravitational equation does. In the present candidate class, matter is not coupled directly to $G_{AB}$, $U^A$, $S$, or $Q^I$. It is coupled only through the bridge metric seen by observers:

$$S^{\text{cand}}_{\text{matter}}[G, U, \Xi, \psi] \equiv S_{\text{matter}}[\Xi^* \mathfrak{g}, \psi] = S_{\text{matter}}[g, \psi]. \tag{25}$$

**Proposition 6.** *If the observer-accessible matter action is of the form* (25)*, then the candidate bridge model has a single universal low-energy matter metric, namely* $g_{\mu\nu}$*.*

*Proof.* All matter variation is taken with respect to the pullback metric $g_{\mu\nu}$. Therefore the matter stress tensor is

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}},$$

and by ordinary diffeomorphism invariance of the matter action one has $\nabla^\mu T_{\mu\nu} = 0$. Since $G_{AB}$, $U^A$, $S$, and $Q^I$ do not appear as independent matter-coupling arguments, there is no second universal matter metric and no direct preferred-frame matter coupling. □

This route excludes, at the candidate level, direct observer-accessible couplings such as

$$J_\mu u^\mu, \qquad \chi(S)\, \mathcal{O}_{\text{matter}}, \qquad q^I \mathcal{O}_I,$$

unless they are driven below the exact-closure threshold. If such terms survive unsuppressed in the observer-accessible regime, then the model no longer belongs to the exact-closure line.

## 9. First Mode-Control Pass

The next task is to say what must happen to the extra substrate sector if the candidate model is to remain answerable to exact closure. For that purpose, linearize about the homogeneous bridge background

$$\bar{G}_{AB} = \delta_{AB}, \qquad \bar{U}^A = \delta^A{}_0, \qquad \bar{S} = \sigma_0 X^0, \qquad \bar{Q}^I = 0. \tag{26}$$

Then

$$\bar{\mathfrak{g}}_{AB} = \text{diag}(-1, 1, 1, 1). \tag{27}$$

Write perturbations as

$$G_{AB} = \delta_{AB} + H_{AB}, \qquad U^A = \bar{U}^A + V^A, \qquad S = \sigma_0 X^0 + \sigma, \qquad Q^I = q^I. \tag{28}$$

The unit constraint (1) then gives at linear order

$$V^0 = -\frac{1}{2} H_{00}. \tag{29}$$

Thus only the three spatial components of $V^A$ remain independent before further decomposition. Writing

$$V^i = V^i_T + \partial^i \chi, \qquad \partial_i V^i_T = 0, \tag{30}$$

one sees that the extra nonmetric sector consists of the scalar pair $(\sigma, \chi)$, the transverse vector $V_T^i$, and the multiplet $q^I$.

The bridge perturbation itself is

$$\delta\mathfrak{g}_{00} = -H_{00}, \qquad \delta\mathfrak{g}_{0i} = -H_{0i} - 2V_i, \qquad \delta\mathfrak{g}_{ij} = H_{ij}. \tag{31}$$

Equation (31) makes the mode-control problem explicit: unless $V_i$, $\sigma$, and $q^I$ are constrained, screened, or gapped, they feed directly into the observer-accessible Lorentzian sector.

The candidate exact-closure branch is therefore defined by the low-energy hierarchy

$$E^2,\ |\mathbf{k}|^2 \ll M_*^2, \qquad M_*^2 \equiv \min\Big\{m_S^2,\ \lambda_{\min}(M),\ M_U^2\Big\}, \tag{32}$$

where $M_U$ denotes the characteristic suppression scale of the nonmetric $U$-sector generated by the constraint structure and the $\kappa_\theta, \kappa_\Omega$ terms. If the $U$-sector is purely auxiliary in a given branch, one may read $M_U = \infty$.

In that regime the scalar and vector sectors may be integrated out, producing only derivative-suppressed corrections,

$$\Delta S[g,\psi] = \int d^4x\,\sqrt{-g}\,\mathcal{O}\left(\frac{\nabla^2}{M_*^2}\right), \tag{33}$$

so that the unsuppressed observer-accessible propagating sector reduces to the usual transverse-traceless tensor modes of the exact relativistic theory. This is not yet a full quadratic proof. It is the first controlled mode statement needed to keep the candidate bridge model scientifically disciplined.

## 10. Exact-Closure Test

With (24), (25), and (32) in hand, one can now state the exact-closure test explicitly. The candidate substrate model remains in the active Æther-flow Interpretation of Relativity line only if all of the following hold in the observer-accessible regime:

1. **Single-metric matter test.** Matter and light couple only to $g_{\mu\nu}$, with no unsuppressed direct coupling to $U^A$, $S$, or $Q^I$.

2. **Field-equation test.** The bridge correction tensor in (24) satisfies

$$\Delta_{\mu\nu}^{(U,S,Q)} = 0 \tag{34}$$

   exactly, or at minimum is suppressed below observational relevance by the gap condition (32).

3. **Spectrum test.** No unsuppressed extra scalar or vector pole survives in the observer-accessible propagator.

4. **Recovery test.** The weak-field, redshift, causal, and local-SR structures match the exact-closure manuscripts already in the active sequence.

If those tests are passed, the candidate model lies on an exact-closure branch of the derivational program. If they fail, the correct scientific status is not to blur the failure with the established theory, but to contract back to the exact-closure benchmark already fixed by the existing manuscripts.

**Remark 3.** *This is the precise sense in which the present paper now does all four missing pieces of*

*the next research step. It states one explicit candidate substrate dynamics, one universal matter-coupling route, one first mode-control regime, and one exact-closure test. What it still does not do is prove that the candidate model actually passes that test.*

## 11. What This Manuscript Establishes and What It Does Not

The present manuscript establishes the following positive structure.

1. It makes the substrate variables explicit through $(\mathcal{A}, G_{AB}, U^A, S, Q)$.
2. It identifies the local experiential three-space with the substrate projector $P_{AB}$ orthogonal to the ordered flow.
3. It gives a concrete Lorentzian bridge metric $\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$.
4. It defines the observer map by which the effective metric $g_{\mu\nu}$, the effective observer field $u^\mu$, and the rest-space metric $h_{\mu\nu}$ are recovered.
5. It proposes one explicit candidate substrate action class (20)–(21).
6. It gives a universal matter-coupling route through $S_{\text{matter}}[g, \psi]$ alone.
7. It states a first controlled mode-gap condition (32) for the extra substrate sector.
8. It states an explicit exact-closure test through (24) and (34).

The present manuscript does *not* establish the following.

1. It does not derive the bridge-sector Einstein term from a deeper still-hidden microphysics.
2. It does not prove that the candidate action actually drives $\Delta_{\mu\nu}^{(U,S,Q)}$ to zero in every admissible regime.
3. It does not complete the full quadratic mode counting or nonlinear stability analysis of the candidate substrate model.
4. It does not introduce any established non-GR low-energy observable signature within Æther-flow Interpretation of Relativity.

**Remark 4.** *The correct scientific status of the manuscript is therefore intermediate but substantive. It closes the missing bridge from ontology to an explicit candidate relativistic sector, while leaving the decisive dynamical proof to later work. Until that later work succeeds, exact closure remains the benchmark and reversion theory of the framework.*

## 12. Conclusion

The project could not advance indefinitely by repeating exact GR / SR recovery statements. What it needed next was a disciplined bridge from ontology to explicit substrate kinematics, candidate dynamics, and a testable route toward derivation. The present manuscript supplies that next layer.

The core move is simple but consequential. The Æther is represented by a four-dimensional substrate manifold with positive-definite metric $G_{AB}$. The Æther-flow is represented kinematically by a unit ordered-flow field $U^A$ together with an order scalar $S$ monotone along that flow. Observed

three-dimensional space is represented by the local projector orthogonal to $U^A$. The observer-accessible Lorentzian sector is then obtained through the bridge metric $\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B$ and its pullback to the effective spacetime metric $g_{\mu\nu}$. On that bridge, the manuscript now adds one candidate action class, one universal matter-coupling route, one first mode-control condition, and one exact-closure test.

This is not yet a completed derivation of Einsteinian gravity. It is the first explicit candidate dynamical architecture on which such a derivation would have to stand. The exact-closure theory of Æther-flow Interpretation of Relativity therefore remains scientifically primary, but the path beyond exact closure is now sharper: the candidate model must either satisfy the exact-closure test stated here or yield no promotion beyond the benchmark theory already fixed by the active sequence.